Major Project – 2 Report

1. Project Title:

Loan Approval Prediction using Machine Learning Classification Techniques.

2. Objective:

The objective of this project is to develop classification models to predict whether a loan application will be approved or rejected based on socio-economic and financial factors such as income, credit history, employment status, loan amount, and property area. The goal is to analyze the key determinants influencing loan approval decisions and build a reliable predictive model that can assist financial institutions in risk assessment and decision-making.

3. Tools and Libraries Used:

Programming Environment:

- Python (Jupyter Notebook)

Libraries Used:

- NumPy – Numerical computations
- Pandas – Data cleaning and preprocessing
- Matplotlib & Seaborn – Data visualization
- Scikit-learn – Model building and evaluation

Classification Models Implemented:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- Support Vector Machine (SVM)

Evaluation Metrics Used:

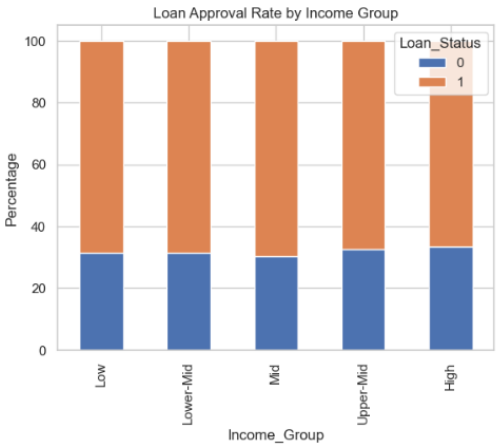
- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix
- ROC-AUC Score

4. Screenshots of Key Visualizations:

1)

2)

3)



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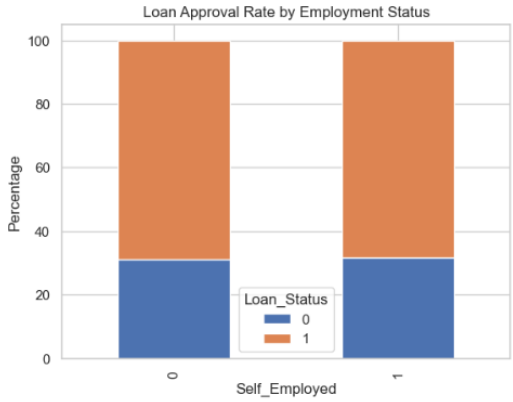

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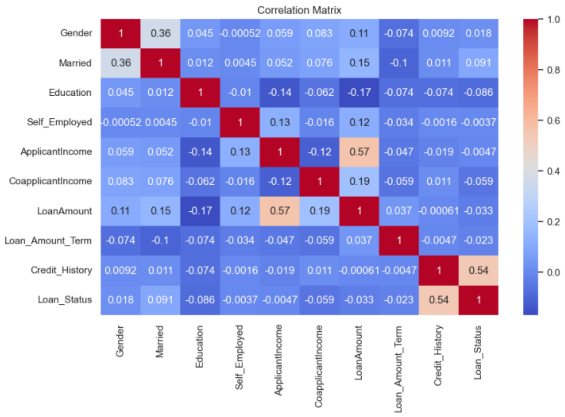

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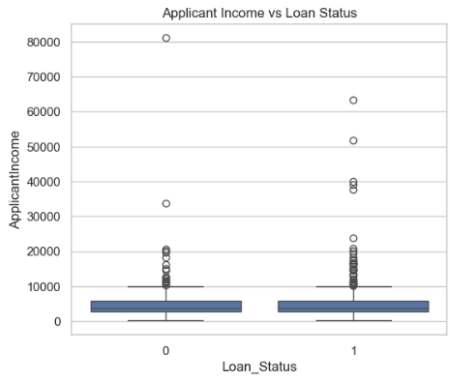

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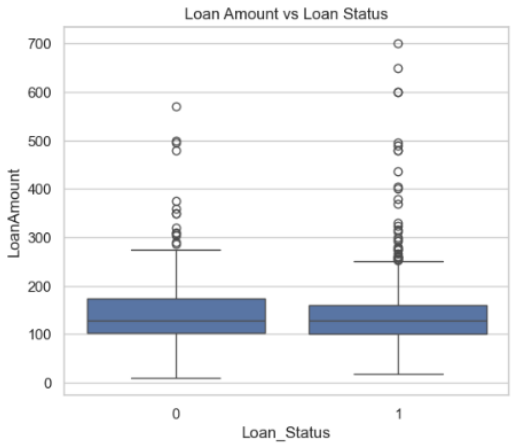

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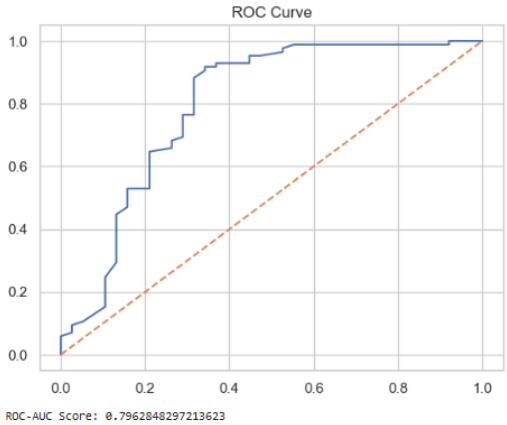

5. Summary of Insights:

Exploratory Data Analysis (EDA):

- Loan approval rates are relatively similar across genders, indicating that gender alone is not a strong predictor.
- Higher income groups show slightly better approval rates, though income alone does not guarantee approval.
- Employment status moderately influences loan approval outcomes.
- Credit history shows a strong positive correlation with loan approval, making it a critical factor.
- Applicants requesting moderate loan amounts tend to have better approval chances compared to very high loan amounts.

Model Building & Evaluation Insights:

- Logistic Regression provided a solid baseline performance for binary classification.
- Decision Tree captured non-linear patterns but showed potential overfitting.
- Random Forest achieved the best overall performance due to its ensemble learning capability and robustness.
- SVM performed well but required proper feature scaling.
- The confusion matrix showed that the model correctly classified the majority of approved and rejected applications.
- A high ROC-AUC score indicated strong class separability and reliable prediction performance.

Business Insights:

- Credit history is the most influential factor in loan approval decisions.
- Income stability and reasonable loan amounts improve approval probability.
- The Random Forest model can be deployed as a decision-support tool for automated loan screening.

- The system can help banks reduce processing time and minimize credit risk.
- Data-driven loan evaluation can improve consistency and reduce human bias.

6. Challenges Faced and Solutions:

1. Handling Missing Values:
Several features contained missing data, which were handled using median imputation for numerical variables and mode imputation for categorical variables.

2. Converting Categorical Variables:
Categorical features were converted into numerical format using Label Encoding and One-Hot Encoding to make them suitable for machine learning models.

3. Special Values in Dependents Column:
The value "3+" caused errors during scaling and was converted into numeric form.

4. Temporary EDA Columns Affecting Modeling:
An income grouping column created for visualization caused scaling errors and was removed before model training.

5. Feature Scaling Issues:
StandardScaler was applied only after ensuring all features were numeric, resolving transformation errors.

6. Model Overfitting:
Decision Tree showed signs of overfitting; Random Forest provided a more generalized solution.

7. How a Bank Would Use This Model in Practice:

In practice, a bank can integrate this model into its loan management system to assist in evaluating loan applications. When a customer applies for a loan, their financial and personal details can be input into the model to generate a probability score for approval.

Applications predicted as low-risk can be fast-tracked, while high-risk applications can be flagged for detailed manual review. This improves operational efficiency, reduces processing time, and enhances credit risk management. The model serves as a decision-support system rather than a replacement for human judgment, enabling data-driven and consistent loan approval processes.

8. Conclusion:

This project successfully demonstrated the application of classification techniques to predict loan approval status using financial and socio-economic data. Through systematic preprocessing, exploratory data analysis, model comparison, and evaluation, meaningful insights were derived about the factors influencing loan decisions.

Among the models tested, Random Forest provided the best predictive performance due to its ability to capture complex feature relationships and reduce overfitting. The project highlights the importance of proper preprocessing, feature selection, and evaluation in building reliable machine learning solutions for real-world financial applications.